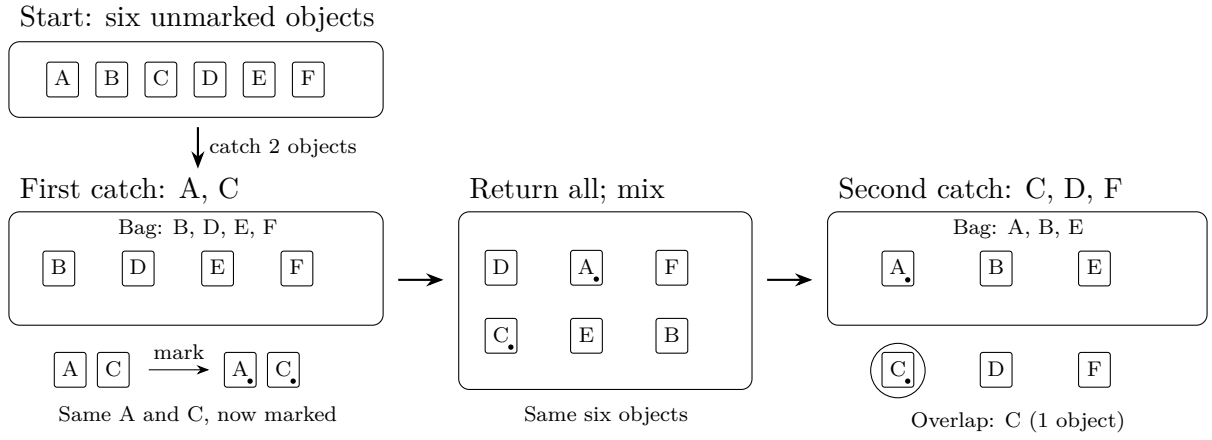
Use identical-size cards or counters with visible marker dots that cannot be felt. Keep the same objects throughout a series; add, remove or exchange none except for temporary catches. For sampling trials, draw without looking; every group of the chosen size should have the same chance, whether marked or unmarked.

Keep each catch out until it is counted. Mark the first catch, return *all* of it, and mix before the second catch. Marks stay on through both catches. An overlap is a marked object in the second catch. Before a new round, return everything and clear the marks. To show a record is possible, you may arrange catches by hand from the same bag; no object appears twice in one catch.



Problem 1: The keeper hides 12 to 24 objects. Catch 6, mark and return them, then catch 6 again. Play three rounds with the same objects. Choose a guess before the keeper reveals the total.

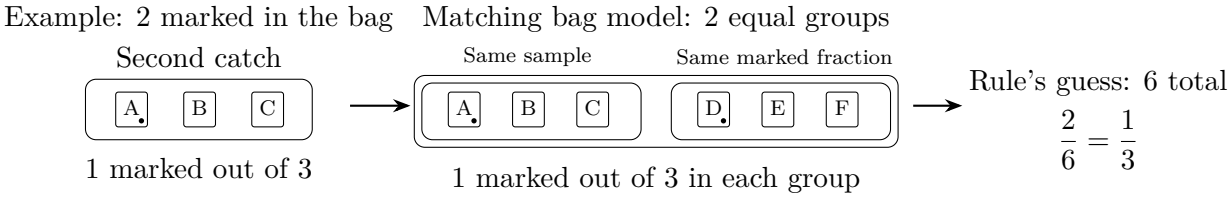
Round	Overlap	My guess
1		
2		
3		

Keeper's total: _____

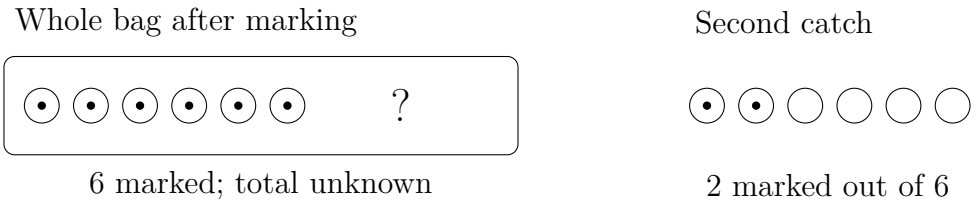
Problem 2: Both records have first catch 6 and second catch 6. For each record, build the smallest possible bag. Is there a largest possible bag if you have as many cards as you want?

Record	Overlap	Smallest total
A	2	
B	0	

Problem 3: A proposed guessing rule chooses a total that makes the marked fraction in the bag match the marked fraction in the second catch.



Find the rule's guess for record A. Does the catch record force that total?



Problem 4: These example rounds all use one unchanged bag, with first catch 6 and second catch 6. Could a bag of 18 objects produce every record? Decide what you would report about the total after seeing all five rounds.

Round	Overlap	Rule's guess
1	1	36
2	2	18
3	3	12
4	2	18
5	0	No matching total

Problem 5: The last round caught no marked objects. The keeper says the bag holds at most 20. Which catch sizes make an overlap of 0 impossible? Explain.

Problem 6: A bag has 24 objects. You may handle 12 objects altogether: first catch size + second catch size = 12. Choose the split with the greatest average overlap over many rounds. Explain why no other split does better.

The average over all equally likely second catches is

$$\text{average overlap} = \frac{\text{first catch size} \times \text{second catch size}}{\text{bag total}}.$$

Problem 7: Would the best split change with a budget of 13 objects? What about 20? Find a rule for any whole-number budget from 2 to 24 and explain why it works.

Problem 8: A keeper says, “Our overlap was small, so our bag must be large.” Some marks might have rubbed off. How could that change the guess? Design a way to test whether lost marks, rather than the total, caused the small overlap.